10.1 Unit description

Introduction

Students are expected to further develop the experimental skills that they acquired in Units 1 and 2.

Students are expected to develop these skills, and a knowledge and understanding of experimental techniques, by carrying out a range of practical experiments and investigations while they study Units 4 and 5.

This unit will assess students' knowledge and understanding of experimental procedures and techniques that were developed when they conducted these experiments.

Development of practical skills, knowledge and understanding

Students should do a variety of practical work during the IA2 course to develop their practical skills.

Centres should provide opportunities for students to plan experiments, implement their plans, collect data, analyse their data and draw conclusions in order to prepare them for the assessment of this unit.

Experiments should cover a range of different topic areas and use of a variety of practical techniques. The specification for Units 4 and 5 contain suggestions for practical work, although these suggestions do not constitute an exhaustive list. This should help students to gain an understanding and knowledge of the practical techniques that are used in experimental work.

Students should gain experience of using log graphs to determine the relationship between two variables. The graphs do not always need to be obtained for variables that are related by the exponential function.

For example, students could investigate how the pressure of a fixed mass of gas varies with its volume at constant temperature and plot an appropriate log/log graph to determine the relationship between the pressure and volume of the gas.

How Science Works

Students should be given the opportunity to develop their practical skills for *How Science Works*, numbers 2–6, as detailed in *Appendix 3*, by completing a range of different experiments that require a variety of different practical techniques throughout the International Advanced Level course.

Students should produce laboratory reports on their experimental work using appropriate scientific, technical and mathematical language, conventions and symbols in order to meet the requirements of *How Science Works*, number 8.